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ASSESSMENT TOOL – 4

NAME: V.Preete

REG NO: 192572004

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COURSE NAME: Cloud Computing and Big Data Analysis

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

ANSWER:

Introduction

During seasonal sales such as Black Friday or Diwali, e-commerce websites receive a sudden increase in visitors. Cloud auto-scaling and load balancing help maintain application performance and availability.

Auto-Scaling

Auto-scaling automatically increases or decreases the number of servers (VMs or containers) based on traffic demand.

Benefits:

- Handles sudden traffic spikes.
- Prevents server overload.
- Reduces infrastructure costs by removing unused resources after traffic decreases.

Load Balancing

A load balancer distributes incoming user requests evenly across multiple servers.

Benefits:

- Prevents any single server from becoming overloaded.
- Improves application performance.
- Increases availability by redirecting traffic if one server fails.

Reducing Downtime and Improving User Experience

- Users experience faster page loading.
- Website remains available during heavy traffic.
- Reduces server crashes and failed transactions.
- Ensures smooth shopping and payment processes.

Role of Monitoring Tools

Monitoring tools continuously track:

- CPU usage
- Memory usage
- Network traffic
- Response time
- Error rates

If resource usage exceeds a threshold (e.g., CPU > 80%), monitoring tools trigger auto-scaling to launch additional servers.

Real-World Example

An online shopping platform like Amazon experiences huge traffic during festive sales.

- Load balancers distribute customer requests across multiple servers.
- Auto-scaling launches additional servers during peak demand.

- Monitoring tools detect increased traffic and automatically adjust resources.
- Customers enjoy uninterrupted shopping and faster checkout.

Conclusion

Cloud auto-scaling, load balancing, and monitoring ensure high availability, minimize downtime, and deliver a smooth user experience during peak traffic periods.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data?Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

ANSWER:

Introduction

Organizations store backup data in the cloud to improve data protection, reliability, and disaster recovery.

Durability and Data Redundancy

Cloud storage providers maintain multiple copies of data across different servers and locations.

Benefits:

- Protects against hardware failures.
- Ensures data is not lost.
- Provides high durability and availability.

Encryption

Encryption converts data into unreadable form.

- Data at Rest: Encrypts stored backup files.
- Data in Transit: Encrypts data while transferring over the internet.

Only authorized users with encryption keys can access the data.

Access Control

Access control ensures only authorized users can access backup data.

Methods include:

- Identity and Access Management (IAM)
- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)

Cost Advantages

Compared to traditional storage:

- No need to purchase expensive storage hardware.
- Pay only for the storage used.
- Lower maintenance and operational costs.
- Easy scalability as backup size grows.

Disaster Recovery Example

A company's primary data center is damaged by a flood.

- Cloud backups remain safe in another region.
- Data is restored quickly.
- Business operations resume with minimal downtime.

Conclusion

Cloud storage offers secure, durable, scalable, and cost-effective backup solutions while supporting rapid disaster recovery.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

ANSWER:

Introduction

A Content Delivery Network (CDN) improves application performance by delivering content from servers located closer to users.

How CDNs Improve Performance

- Frequently accessed content is cached on edge servers.
- Users download data from the nearest server instead of the main data center.
- Reduces network congestion.
- Improves page loading speed.

Role of Edge Servers

Edge servers:

- Store cached content near users.
- Reduce latency.
- Handle local traffic efficiently.
- Improve application responsiveness.

Global Availability

Cloud providers maintain data centers around the world.

Benefits include**:**

- Automatic routing to the nearest server.
- High availability.
- Failover if one region becomes unavailable.
- Better performance worldwide.

Real-Time Streaming or Gaming Example

A video streaming platform like Netflix uses CDNs to deliver movies from nearby edge servers.
Similarly, an online multiplayer game delivers updates and game content through edge servers, reducing lag and improving gameplay.

Conclusion

CDNs and edge servers reduce latency, improve performance, and provide a consistent user experience for global applications.

4. A financial institution requires strict compliance and data security in cloud usage.Explain how cloud providers support

regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

ANSWER:

Introduction

Financial institutions must protect sensitive customer data while complying with government regulations.

Regulatory Compliance

Cloud providers support standards such as:

- PCI DSS
- ISO 27001
- SOC 2
- GDPR (where applicable)

These certifications help organizations meet legal and security requirements.

Logging and Monitoring

Logging records:

- User logins
- File access
- Configuration changes
- Security events

Monitoring detects:

- Suspicious activities
- Unauthorized access
- System failures
- Security threats

Encryption

Encryption protects:

- Stored financial records.
- Data transmitted over networks.

Even if data is intercepted, it cannot be read without the correct encryption keys.

Data Sovereignty

Data sovereignty means data must remain within a specific country or region.

Organizations can:

- Select cloud regions where data is stored.
- Restrict cross-border data transfers.
- Follow local government regulations.

Example

A bank processes online fund transfers.

- Customer data is encrypted.
- Every transaction is logged.
- Monitoring detects fraudulent activities.
- Data is stored only in approved regional data centers to meet regulatory requirements.

Conclusion

Compliance certifications, logging, monitoring, encryption, and regional data storage help financial institutions securely use cloud services while meeting legal requirements.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

ANSWER:

Introduction

Containerization packages an application with all its dependencies, making it easy to deploy consistently across different environments.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Operating System	Share the host OS	Each VM has its own OS
Startup Time	Seconds	Minutes
Performance	Faster	Slower
Resource Usage	Lightweight	Uses more CPU and memory
Portability	High	Moderate

Kubernetes Orchestration

Kubernetes automates:

- Application deployment
- Scaling
- Load balancing
- Self-healing (restarts failed containers)
- Rolling updates without downtime

Portability Across Clouds

Containers run consistently on:

- Public cloud
- Private cloud
- Hybrid cloud
- On-premises servers

This reduces vendor lock-in and simplifies migration between cloud providers.

Microservices Example

An online shopping application can be divided into separate containerized services:

- User Service
- Product Catalog Service
- Shopping Cart Service
- Payment Service
- Order Service
- Notification Service

Each service runs independently and can be updated or scaled without affecting the others.

Conclusion

Containers provide faster deployment, lower resource usage, and excellent portability. Kubernetes simplifies management and scaling, making containerization ideal for modern cloud-native microservices applications.